HOLLAND ENGINEERING, INC.

Salem Main Line 2 Project Report

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View Interactive GIS Map

Project-provided URL; live availability has not been checked.

A. PROJECT OVERVIEW

The Salem Mainline 2 Pipeline Project includes the installation and surveying of Mainline 2 and its associated pipeline connections and facilities.

This interactive GIS map provides survey data collected throughout the project, including installed pipeline alignments, weld locations, pipe and fitting information, and other surveyed features. The map serves as a centralized reference for reviewing pipeline construction, as-built survey information, and associated project documentation.

Understanding the GIS Map

GIS Map Legend & Navigation Guide

- **Red Points – Survey Locations:** Points collected using survey-grade GPS equipment. Click individual points to view available information, including weld/XR numbers, MTR documentation, inspection photos, pipe specifications, grade cover, and other survey attributes.
- **Red Solid Line – Installed Pipeline:** Represents the surveyed location of the installed pipeline.
- **Orange Dashed Line – Proposed Pipeline:** Represents the proposed pipeline alignment for comparison with the installed location.
- **Gray Lines – Abandoned Pipeline:** Identifies existing pipeline sections that have been abandoned, where applicable.

Map Navigation Tools

- **Location Tool (Map Icon):** Displays your current location on the map when location services are enabled, particularly useful when navigating the project on site.
- **Measurement Tool:** Allows users to measure distances directly on the map.
- **Search Tool (Binoculars Icon):** Allows users to search for specific survey points or weld identifiers, including XR numbers.

Select individual survey points to access available project records and supporting documentation.

B. CONTRACTOR WELDING SUMMARY

Official project statistic	Count	Contractor source
Total welds made	307	Weld Log!H9
Girth welds made	289	Weld Log!I9
Girth welds NDT inspected	289	Weld Log!J9
Girth welds rejected	2	Weld Log!K9
Welds repaired or cut out (combined)	18	Weld Log!M9
Temporary welds made (separate)	19	TW Log!E8
Temporary welds removed	18	TW Log!F8

Official totals are not replaced by suffix counts or added across temporary/permanent categories. Repaired/cutout is a combined total, not separate repair and cutout counts.

Identifier-derived category	Unique count	Identifiers / notes
All weld identifiers (excluding temporary)	306	Identifiers omitted; enable detailed lists

Identifier-derived category	Unique count	Identifiers / notes
Girth weld identifiers	288	Identifiers omitted; enable detailed lists
NDT inspected identifiers	306	Identifiers omitted; enable detailed lists
Rejected identifiers	2	XR-110, XR-156
Repaired original welds	1	XR-110
Repair identifiers	1	XR-110R
Original welds marked cut out	17	Identifiers omitted; enable detailed lists
Replacement identifiers (N suffix)	7	XR-023N, XR-077N, XR-088N, XR-156N, XR-156NN, XR-207N, XR-208N
Tie-in identifiers	4	XR-TI-001, XR-TI-002, XR-TI-003, XR-TI-004
Intermediate identifiers	2	XR-IW-001, XR-IW-002
Magnetic-particle identifiers	18	Identifiers omitted; enable detailed lists
Temporary weld identifiers	20	Identifiers omitted; enable detailed lists

Identifier counts are deduplicated observations, not official construction totals. Repair Log initial and repair identifiers are checked independently; N-suffix replacements do not establish how many original welds were cut out. Rows marked NOT USED are excluded.

Repair references: XR-110 -> XR-110R (Repair Log!11)

C. PIPELINE 2D STATIONING AND LENGTHS

Total reported 2D footage: 4,858.25 ft | Pipeline sections: 10 | Sections with stationing exceptions: 1

Verified footage: 4,814.93 ft | Provisional footage: 43.32 ft. Provisional spans are marked *. Totals sum the table's reported 2D station spans; 3D pipe measurements are not included.

Pipeline	Beginning	Ending	2D Length (ft)
16in SALEM ML 2	-0+25.40	41+18.91	4,144.31
16in S-2S-2	0+00.00	0+43.32	43.32 *
4in PUSH GAS S-ML2	0+00.00	0+80.93	80.93
6in SALEM LAT-1	0+00.00	0+41.36	41.36
SALEM-WL-316	0+00.00	0+66.71	66.71
10in N-ML2	0+00.00	1+05.63	105.63
4in PUSH GAS N-ML2	0+00.00	1+32.82	132.82
4in 4GSV4	0+00.00	0+15.21	15.21
8in ML3XML2	0+00.00	0+86.98	86.98
8in S-ML3	-0+18.94	1+22.04	140.98

Survey Data Review Notes

16in SALEM ML 2: Excluded separately labeled cut-point/reference section beginning at row 191. Unstationed reference/survey points omitted: rows 36, 37, 38, 40, 41, 105.

16in S-2S-2: * Provisional span; unresolved stationing issues. 2D continuity not confirmed at rows 16-17. Station reset/reversal at rows 17-18. 2D continuity not confirmed at rows 17-18. 2D continuity not confirmed at rows 18-19.

MISSING SURVEYED INSPECTION PHOTOS

Unique identifiers: 15. Checked against actual final staged website files within HEI surveyed weld scope.

MT-007, MT-022, XR-276, XR-277, XR-278, XR-279, XR-282, XR-283, XR-284, XR-285, XR-286, XR-288, XR-290, XR-291, XR-TI-004

MISSING SURVEYED MTR PDFS

Unique identifiers: 4. Checked against actual final staged website files within HEI surveyed weld scope.

109710, 2236167, 325163, B416740